

Article

Effects of Standardized Innovation Management Systems on Innovation Ambidexterity and Innovation Performance

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Abstract: The goal of this research is to investigate the effects of standardized innovation management systems (SIMSs) on innovation ambidexterity and innovation performance and further on innovation capability. A survey was conducted in Türkiye with 178 manufacturing companies. The PLS-SEM method was used to test the hypotheses by using SmartPLS 4.1.0.2 software. The results show that SIMSs have a positive effect on innovation ambidexterity, innovation capability, and innovation performance. In addition, the results confirm a positive relationship between innovation capability and innovation performance, innovation ambidexterity, and innovation performance. According to research results, companies should follow ISO 56002 principles as a standardized innovation management system to achieve innovation ambidexterity, innovation capability, and innovation performance. This is the first known paper investigating the effect of SIMSs on innovation ambidexterity empirically. This research has provided a new link between the newly developing SIMSs and the innovation ambidexterity literature. Therefore, this study contributes to the development of the SIMSs from this perspective.

Keywords: standardized innovation management system; innovation ambidexterity; innovation performance; ISO 56002; innovation capability



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1. Introduction

In today's fast-changing market conditions, there is a growing realization that innovation becomes one of the key elements toward organizational success and competitive advantage [1]. Businesses that recognize this imperative increasingly turn toward standardized innovation management systems (SIMSs) to drive and nurture innovation throughout their organizations [2]. SIMSs holistically manage innovation from the ideation stage to implementation, which helps streamline and optimize the innovation process [3]. However, the effectiveness of these systems on innovation ambidexterity and innovation performance continually raises substantial debate and analysis among academics.

SIMSs aim to systematize innovation activities by providing structured frameworks [4]. They go beyond the traditional focus on research and development (R&D) departments by covering various types of innovation, including product, marketing, organizational, and process innovation [5]. SIMSs facilitate the translation of innovative concepts into tangible products, services, processes, and business models by providing guidelines, methods, and organizational support [6].

The creation of innovation management standards further highlights the importance of standardization in promoting innovation inside organizations [7]. These standards are

structured frameworks, such as ISO 56002 [8], that enable the benchmarking and optimization of the respective company's innovation processes [6]. Research has shown that organizations that follow innovation management standards achieve significant improvements in various areas, including knowledge management, innovation culture, and the application of innovation tools [9,10]. Therefore, the implementation of SIMSs is likely to help the organization improve its innovation performance [4] and create a continuous culture of innovation [9].

The previous literature on this topic has mostly focused on case studies from different sectors regarding the implementation of SIMSs [5,11–13]. However, empirical studies are relatively limited. Among these, Mir et al. [4] examined the impact of SIMSs on innovation capability (IC) and innovation performance (IP), while Martinez-Costa et al. [7] elaborated upon their effect on product, process, marketing, and organizational innovations. Giménez Espin et al. [2] specifically looked at the influence of SIMSs on radical and incremental innovation. All these studies utilized the UNE 166002 standard [14] as a benchmark and sampled firms from Spain. On the other hand, the only empirical study in the literature about standards and innovation ambidexterity (IA) was conducted by Pertusa-Ortega [15], which focused specifically on quality standards. Apart from this, there are two conceptual studies about the relationships between quality standards and IA [16,17]. As far as we know, none of the studies in the literature have empirically investigated the impact of SIMSs on IA.

Therefore, to address this gap in the literature, this present study sets its objective to explore the effect of SIMSs on IA, IC, and further on IP. Based on dynamic capabilities theory (DCT), this research will seek to provide insights into how organizations can effectively use innovation management standards to enhance ambidextrous strategies and their innovation capabilities toward recording high IP with respect to attaining sustainable competitive advantages. DCT suggests that firms need dynamic capabilities, which enable adaptation to change in the market and structuring organizational resources accordingly [18], to differentiate themselves from rivals and boost performance outcomes [19]. Since SIMSs are considered a dynamic capability [4], they may influence IP. On the other hand, He et al. [20] remarked that dynamic capabilities improve firms' capabilities to cope with environmental uncertainty. They also indicated that dynamic capabilities may support exploratory innovation by focusing on the environment, improving the quality of exploitative innovations, and enabling the balance between exploratory and exploitative innovation. Moreover, Wen and Wen [21] suggest that as dynamic capabilities expand the organizational resource base to realize opportunities, they may develop a firm's innovative capabilities. Therefore, SIMSs may affect IA and IC. Additionally, both IA [22] and IC [23] are considered dynamic capabilities. IA enables firms to simultaneously exploit two capabilities to adapt and respond to changing environmental conditions, thus overcoming potential problems arising from focusing on a single innovation and improving performance outcomes [24]. IC is also effective in all innovation practices and its outcome [21]. Thus, IA and IC may improve innovation performance. That is, drawing on DCT, the main goal of this research is to respond to the following research questions:

RQ1: Do SIMSs have an impact on IA?

RQ2: Do SIMSs have an impact on IC?

RQ3: Do SIMSs improve IP?

The primary contribution introduced by this study to the literature is the empirical investigation of the impact of SIMSs on IA for the first time. A secondary contribution of this research is the examination of SIMS-IC and SIMS-IP relationships using a sample from Türkiye, which is in the context of the developing country category. Since the data collection for previous studies was carried out in Spain, this study allows the confirmation

and generalization of previous findings reported by Mir et al. [4] and Martinez-Costa et al. [7]. As a third contribution, this study marks the first instance in the literature where all aspects of the ISO 56002 standard principles are employed to measure the SIMS construct. The dimensions used to underpin the measures of the SIMS construct according to UNE 166002 were those of Giménez Espin et al. [2] and Martinez-Costa et al. [7]. However, they did not sufficiently cover important principles of the ISO 56002 standard. Factors of SIMSs such as leadership, collaborations, innovation processes, innovation strategy and policy, and management of innovation portfolios were not included in these studies. In this study, which comprehensively addresses all aspects of ISO 56002 principles, a more detailed investigation into the interplays of SIMSs, IC, IA, and IP is conducted.

The rest of the article is structured as follows. Section 2 provides a literature review, Section 3 presents hypothesis development, and Section 4 discusses methodology. Section 5 provides a discussion of the analyses and results. Section 6 summarizes the discussion and the conclusion, discusses the study findings, highlights the limitations of the study, and suggests some possible areas for future research.

2. Literature Review

2.1. Standardized Innovation Management Systems

Innovation should be a holistic activity that covers the entire company from the top to the bottom unit. Furthermore, innovation must be supported by systems, organizational structures, and a company culture that incentivizes transformative ideas and products [3]. Since innovation is a process, it is important to systematize innovation through the implementation of process management principles [25].

Previous studies on innovation management have predominantly focused on R&D departments with a narrow perspective [4]. Innovation management does not only focus on technological innovations and the implementation of new products. It adopts a comprehensive approach that encompasses the entire innovation process, starting with the creation of an innovation culture to foster the generation of novel ideas [7]. It then extends to managing new ideas, establishing collaborations and innovation ecosystems, developing business processes and models, and considering the human factor that will make these possible [26].

The implementation of innovation management systems will enable companies to achieve long-term success in their innovation endeavors [27]. Consequently, companies will be able to systematically direct the necessary corporate resources for innovation processes in a planned and organized manner [7].

According to Pellicer et al. [25], systematization is positive for innovation, and innovation and standardization are complementary concepts. Additionally, innovation management can be standardized alongside other processes, such as quality and environmental processes [25]. De Vries and Verhagen [28] found in their study that there is a positive correlation between normalization and innovation. Similarly, Mir and Casadesus [5] have emphasized the necessity of a regulatory framework that will include the creation of guidelines and methods for know-how and documentation management in the innovation process.

Conversely, some scholars argue that standardization may impede the creative aspects essential for innovation [29]. However, it is crucial to note that invention, where creativity is necessary, is just one of the steps in the innovation process [30]. Through standardization, it is aimed to institutionalize the innovation culture in organizations that will contribute to the formation of a creative environment [10].

The first innovation management standard, UNE 166002, was published by AENOR in 2002 [7]. This standard was inspired by the ISO 9001 [31] quality standard [5,32]. Subsequently, Portugal (NP 4456-61) [33], Germany (PAS 1073:2008) [34], the U.K. (2008 BS 7000-1:2008) [35], Ireland (NWA 1:2009) [36], Denmark (DS-hæfte 36:2010) [37], and France

(2013 FD X50-271:2013) [38] have developed their innovation management standards [6]. In 2013, the European Committee for Standardization (CEN) created the CEN/TS 16555-1:2013 [39] innovation management standard [2]. In 2019, the ISO innovation management committee (ISO/TC 279) published the ISO 56002 innovation management standard [40].

Companies should assess innovation management standards as a strategic management tool that provides the opportunity to learn from leading practices in this field and to apply already tested innovation management approaches and methodologies [9].

Studies conducted on companies that obtained the innovation management system standard certification in Portugal (NP 4457:2007, 200 companies) and in Spain (UNE 166002, 280 companies) showed that these firms achieved significant advancements in areas such as knowledge management, the establishment of an innovation culture, monitoring and documentation of innovation processes, creativity and idea generation, and the application of innovation tools and methods after the standardization of innovation management [9,10].

In summary, the innovation literature emphasizes the necessity of the topics covered in the standard to support SIMSs. The SIMSs recommended by the standard will help companies execute their innovation processes more efficiently [7]. SIMSs are expected to accelerate the dissemination of innovation and reduce risks, thereby decreasing the time to market for innovations [41].

2.2. Innovation Capability

The IC of a company is its ability to continuously develop innovations by transforming the knowledge and ideas it generates and acquires into new products, processes, and services [42]. IC enables the company to meet rapidly changing current and future customer expectations and achieve success in market conditions by creating value for the benefit of the company and its stakeholders [43].

Laforet [44] emphasized that innovation can only occur in companies with the capability for innovation. Samson et al. [45] underscored that IC, a prerequisite for value creation, helps to trigger innovation by playing an important role in shaping and managing companies' diverse capabilities. Consequently, this enables firms to develop new products faster, implement appropriate process technologies, meet market expectations, and mitigate competitive threats [46].

The relationship between IC and competition is appropriately addressed through the lens of dynamic capability theory [4]. Companies must acquire internal and external competencies and capabilities to adapt to rapidly changing environmental conditions, gain a competitive advantage, and sustain it. This organizational capability is defined as dynamic capability [18]. Eisenhardt and Martin [18] defined dynamic capability as specific and identifiable processes and routines. Through these processes, dynamic capability becomes an important source of the organization's ability to create value [45]. SIMSs and IC can be considered dynamic capabilities because they provide companies with new innovation-related competencies and capabilities, thus enabling adaptation and competitive advantage in changing environmental conditions [42].

Numerous studies in the literature have explored common characteristics of innovative companies. Based on these studies, factors influencing companies' ability to manage innovation processes have been identified as components of IC [42,47]. There is no common agreement on these factors. These factors are not independent but interact with each other [48]. IC comprises various practices and processes within the company that trigger and enhance innovation [42]. Some of the factors influencing IC defined in the literature are leadership, organizational culture, customer focus, external knowledge utilization, orientation to change, and creativity of employees [45,47].

In the literature, IC is conceptualized in two different ways: as an output and a process [49]. Some scholars, such as Menguc and Auh [50], have studied IC as a one-dimensional structure consisting of product and process innovation output. In these studies, IC dimensions such as behavior changes and processes were neglected [51]. While some academics who conceptualize IC as a process used a unidimensional structure [43,52], others measured IC with a multidimensional structure [45,47].

2.3. Innovation Ambidexterity

Ambidextrous organizations are more successful both in exploitative innovation by improving their existing competencies and in seeking new opportunities that will enable explorative innovation [53]. Thus, they meet the current expectations of their customers, use the opportunities in the current market, and at the same time, they are prepared for the challenging market conditions and new markets in the future [54]. Companies need to enhance their revenues in the current market through exploitative innovation activities and direct their revenues to explorative innovation activities that will enable new technologies and new markets for the sustainability of the company in the future [55]. Achieving a balance between explorative and exploratory innovation, which compete for the use of company resources, is vital for companies to perform successfully now and in the future [56].

Exploitative innovation encompasses the development of current products, processes, markets, and organizational structures to meet existing customer expectations and market conditions by improving existing competencies and technologies [57]. On the other hand, explorative innovation involves creating completely new products, processes, organizations, and markets through new competencies and technologies to meet new customer expectations in new markets [58]. Both exploitative and explorative innovation contribute to acquiring new knowledge despite differing in type and extent [59]. As March [60] stated, while exploitative innovation directs its focus toward improvement and efficiency, explorative innovation centers on research, discovery, and experimentation. Hence, exploratory activities involve uncertainty in returns, while the returns from exploitative activities are more predictable [60]. Exploitative innovation focuses on short-term successes without emphasizing long-term sustainability, whereas explorative innovation concentrates on long-term achievements, neglecting short-term outputs [57].

When companies focus only on exploitative innovation, their short-term success in the current market and competition can cause companies to fall into the “success trap”. In this situation, where companies ignore explorative innovation activities, their existing core capabilities may become stronger and turn into rigidities that will make it difficult for companies to adapt to future expectations and market conditions. When market conditions shift, these strong competencies may become invalid in new market conditions. This situation poses a serious risk to the sustainability of companies [55].

Likewise, when the company focuses only on exploratory activities, failures that may occur while searching for different opportunities and trying different alternatives before receiving the financial return of the exploratory innovation activities may pull the company into the “failure trap”. In this case, since there are no exploitative innovation activities that will ensure their continuous success in the current market and provide cash flow, serious problems may occur for both the continuation of the exploratory activities and the sustainability of the company [55].

Several researchers have conceptualized IA as a complex dynamic capability that allows companies to simultaneously balance different and conflicting explorative and exploitative innovation activities [61,62]. This dynamic capability involves processes and routines that facilitate the integration and coordination of conflicting exploratory and

exploitative activities [19,61,62]. IA is a critical dynamic capability that can give companies a competitive advantage [54].

In the literature on IA, scholars have different approaches to conceptualizing explorative and exploitative innovation, and no consensus has been reached on this issue [59]. March [60] defines explorative and exploitative innovation as two opposite points of a continuous line and emphasizes that explorative and exploitative innovation compete for company resources. Furthermore, March [60] highlights the importance of the balance dimension to achieve superior company performance [54]. Conversely, some scholars argue that explorative and exploitative innovation are complementary and independent of each other, underlining that companies should be at high levels of both explorative and exploitative innovation and consider the combined dimension of ambidexterity [53,56,63,64]. Another group perceives IA as a higher-order construct [64]. At least some academics also foresee IA as a single structure [65].

3. Hypothesis Development

3.1. SIMSs and Innovation Capability

Contrary to the view that standards hinder innovation, standards ensure coordination and integration in the innovation process. Especially international standards, which are formed with the consensus of academics and experts from different universities and organizations around the world, enable companies to significantly increase their innovation ability with the best practices and technical knowledge in the industry. In parallel to this, studies in the literature also acknowledge that there is a positive relationship between standards and innovation [41].

Certain aspects contributing to IC are also found within the framework of innovation management system standards. Consequently, companies will concurrently enhance their innovation capabilities by adhering to innovation management standards [47].

Innovation management is embraced as a structured process rather than a well-intentioned and singular strategy [66]. To achieve success in their innovation endeavors, companies need to manage interactive and interrelated innovation activities and resources with a holistic and systematic approach [32]. Implementation of SIMSs provides companies with opportunities to manage innovation processes more effectively and continuously improve them [4].

Ukko et al. [67] demonstrated that companies can enhance innovation capabilities by utilizing structured processes such as innovation management standards. Research conducted on a Spanish manufacturing firm revealed that the implementation of the UNE 166002 innovation management system standard resulted in the strengthening of the innovation culture, improvement in decision-making mechanisms, enhancement of the company's image, and an increase in the company's IC [5]. In a more extensive empirical study with 373 companies, Mir et al. [4] have identified that the standardization of innovation management systems (UNE 166002) had a substantial and favorable impact on IC.

In parallel with previous studies mentioned above, we propose the following hypothesis:

H1: *Standardized innovation management systems are positively associated with innovation capability.*

3.2. Innovation Capability and Innovation Performance

IC enables the improvement of a company's IP through the conversion of new ideas and knowledge into products, processes, organizational, and marketing practices, which leads to a competitive advantage in the market [68,69]. Consequently, the improvement of companies' IP is contingent upon their possession of IC [69]. A company's IP will be

more effective to the extent that its IC is stronger, as evidenced by studies conducted by Mir et al. [4] and Lawson and Samson [42].

Samson et al. [45] defined IC from a holistic viewpoint as several core elements needing to synergize for successful outcomes in innovations. Samson et al. [45] found that five of the eight components directly influenced the innovation performance in their research. These variables will consist of project portfolio management to balance the innovation projects, generating value from the innovation project, removing non-value-adding projects, reducing risk potential values, and improving processes of the idea generation. Saunila [47] established that participative leadership, idea generation, organizational structures, and know-how elements had the greatest influences on performance. In another study, Mir et al. [40] claim that the market, organizational structure, strategy, and network dimensions of IC positively affect IP. More recently, Sarwar et al. [69] and Hurtado-Palomino et al. [52] also found a positive relationship between IC and IP, in parallel with the results of studies conducted in previous years.

Therefore, in line with the results of the existing literature, we posit the following hypothesis:

H2: *Innovation capability is positively associated with innovation performance.*

3.3. SIMSs and Innovation Ambidexterity

Formalization refers to the explicit definition of rules, procedures, instructions, and standards pertaining to business processes [70]. Due to the neglect of supportive and coercive formalization types in some literature studies, diverse findings have emerged regarding the correlation between formalization and innovation [71]. Coercive formalization seeks to enforce compliance with these procedures and standards among employees, whereas supportive formalization aims to influence employees in a motivating, guiding, and facilitating direction [72]. The ISO 56002 standard, as outlined in Clause 1, is aimed at providing guidance for innovation management, indicating its supportive nature. Supportive, guiding formalization stimulates IA by promoting knowledge formation and creativity [73].

Delving into the ISO 56002 innovation management standard's perspective on IA (Section 0.4), it is highlighted that this standard will aid organizations in achieving a balance between exploratory and exploitative innovation activities. Furthermore, IA is promoted in the different sections of the standard, such as knowledge management, innovation portfolio management, and organizational structures.

Innovation ambidexterity refers to the necessity for companies to balance exploratory and exploitative innovation activities to achieve sustainable growth [60]. Exploitative innovation emphasizes incremental innovation, whereas exploratory innovation focuses on radical innovation [15]. The ISO 56002 standard, in Sections 1.2b, 4.2.2h, and 4.2.2d, discusses the applicability of this standard to various types of innovation, ranging from incremental innovation to radical innovation. As highlighted in several studies, the standardization of innovation processes facilitates the development of different types of innovation, from incremental to radical innovation [74,75].

ISO 56002 innovation management standard emphasizes in Section 7.1.4 that companies should create knowledge management approaches for an effective innovation management system. Knowledge management capability comprises structures that facilitate the accumulation of organizational knowledge and its conversion into applicable forms [76]. Knowledge management capability enables organizations to collect and benefit from ideas about new products, new trends, market, and competition issues by constantly interacting with their employees [77]. It also enables them to obtain information through collaborations with organizations [78]. In this respect, knowledge management capability is extremely useful for both explorative and exploitative innovation [77]. Ambidextrous organizations

concentrate on improving existing knowledge, methods, and practices to encourage incremental innovation while exploring new knowledge, methods, and practices to foster radical innovation [54]. Organizations exhibiting advanced knowledge management capabilities demonstrate greater success in fostering IA [78].

Portfolio management methodology, which adopts a holistic view in selecting innovation projects, provides organizations with effective foresight to manage IA [79]. As mentioned in Section 6.4 of the ISO 56002 standard, organizations should develop their innovation portfolios based on appropriate risk and return expectations, types of innovation (incremental and radical), and the level of innovation novelty. As stated in Section 6.4, organizations can construct their innovation portfolios as a combination of projects focused on improving existing products/processes and new products/processes for new customers and markets. Portfolio management thus supports balancing incremental and radical innovation projects, enabling IA [80].

ISO 56002 standard, in the organizational structures Section 6.3, underscores that senior management should consider the structures in which explorative and exploitative innovation activities can be carried out simultaneously or integrated within the organization. It also states that dedicated structures, that is, structural ambidexterity, should be evaluated in cases that require a different leadership style and innovation culture, where uncertainty is high and radical innovation is aimed compared to existing products.

Given the rationales outlined above, we formulate the following hypothesis:

H3: *Standardized innovation management systems are positively associated with innovation ambidexterity.*

3.4. SIMSs and Innovation Performance

From a broad perspective, especially international standards, since they are formed with the participation of many experts in developed countries, contain innovation and excellence that include the best practices, processes, and knowledge of the best companies and institutions in developed countries [41]. Thanks to standards, the process, technology, and norms used by a company or an institution can be used by other companies, contributing to the diffusion of innovation [81]. The implementation of different standards (ISO 9000 [82], ISO 14000 [83], etc.) provides the opportunity to improve companies' processes and procedures [84]. Consequently, it is a natural result that standards inspire the companies' processes and organizational innovation efforts [2,84]. Therefore, the standards, due to their characteristics that trigger change and innovation, create a positive impact on the effective management of processes and successful IP [41].

According to Karlsson and Magnusson [32], the innovation process encompasses interacting complex components such as competencies, organizational structures, evaluation methods, top management support, and collaborations. Successful innovation outcomes are achievable only through the effective and comprehensive management of this process [32].

The SIMSs facilitate the emergence of various types of innovation by supporting collaborations with other companies and institutions, promoting an innovation culture, providing necessary resources for innovation projects, managing the innovation project portfolio, and supporting the formation of new knowledge [2]. It also enables better planning, tracking, and management of innovation projects [5].

SIMSs not only ensure that companies keep track of changes in markets, technology, and customer demands but also help them acquire innovation capabilities and competencies [19]. Hence, SIMSs are also considered a kind of dynamic capability [42]. The innovation management systems standard adopts a dynamic approach rather than a static one. It is possible for companies to take the necessary corrective actions by regularly evaluating processes with Deming's PDCA principle in the innovation management systems

standard. SIMSs, published to streamline innovation processes, also foster rapid adaptation owing to their dynamic nature [4].

ISO 56002 promotes the regular analysis of internal factors within the company and its external environment, including the market, competitors, legal regulations, and economic, social, cultural, and political context. Additionally, it encourages the understanding of the expectations of potential and existing customers, stakeholders, regulatory authorities, and all relevant parties. Furthermore, it encourages collaborations as an external source of knowledge. From this perspective, innovation management system standards can be considered a form of organizational learning system [4]. As Jiménez-Jiménez and Sanz-Valle [85] found in their research, organizational learning systems improve innovation performance [85].

As outlined from various perspectives above, we posit the following hypothesis:

H4: *Innovation management system standards exhibit a positive association with innovation performance.*

3.5. Innovation Ambidexterity and Innovation Performance

Several academics bring attention to the competition between exploratory and exploitative innovation for organizational resources and advocate for the balance of exploitative and exploratory innovation activities, emphasizing the balance dimension of IA [53,56]; conversely, other scholars stress the mutual synergy between exploitative and explorative innovation, highlighting the combined dimension of IA [63].

Exploitative and explorative innovation need not always be in competition; rather, they can mutually reinforce each other. These innovation endeavors might be dispersed across different domains within the organization, thus avoiding the need to compete for the same resources [59]. Success in a product or technology domain via exploratory innovation can have a positive influence on complementary exploitative innovation activities in another domain. Accordingly, explorative innovation may improve the short-term IP through exploitative innovation [53,54]. Because of their different nature, both exploitative and explorative innovation demand different resources, structures, and capabilities that influence different dimensions of IP [62]. Exploitative innovation supports static optimization, whereas explorative innovation promotes dynamic optimization. While strengthening and leveraging an organization's existing competencies are vital in a short-term stable environment, the discovery and acquisition of new competencies become increasingly crucial in the long term [86].

The new knowledge, resources, and competencies that organizations acquire during exploratory innovation activities can subsequently benefit exploitative innovation endeavors. As a result, explorative innovation may indirectly facilitate exploitative innovation [87]. Likewise, the knowledge and competencies necessary for explorative innovation can be attained through exploitative innovation practices [88]. This circumstance results in the significant influence of organizations' incremental (exploitative) IP on their radical (exploratory) IP [89].

As a result, organizations that simultaneously engage in explorative and exploitative innovation activities will achieve more efficient utilization of existing resources and enhance the effectiveness of combining these two types of innovation activities. Consequently, IA will unveil innovation synergies through the interactions between explorative and exploitative innovation [87]. Therefore, we posit the following hypothesis:

H5: *Innovation ambidexterity positively associated with innovation performance.*

Figure 1 shows the research model, which covers all the hypotheses raised in this study.

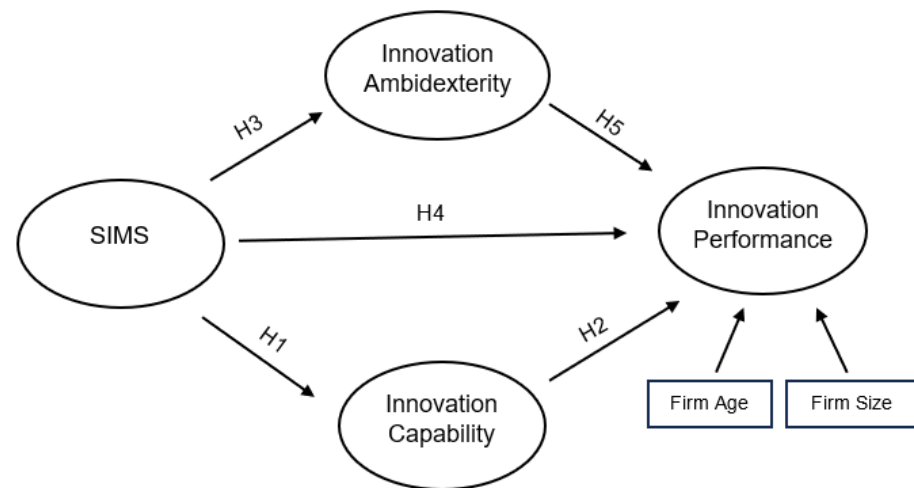


Figure 1. Theoretical model.

4. Methodology

4.1. Measures

All constructs in this study, excluding SIMSs, have been derived from scales utilized in prior research. All items used to measure the constructs were assessed using a 7-point Likert scale.

In measuring IP, a seven-item scale developed by Gunday et al. [1] was utilized. In this research, IC was assessed as a one-dimensional construct. Five items adapted from Akman and Yilmaz [43] were used to measure IC as utilized by various scholars [52,68,69].

Explorative innovation was assessed with five items constructed by Lubatkin et al. [64] and He and Wong [56]. Meanwhile, exploitative innovation was measured using seven items adapted from Jansen [58] and Lubatkin et al. [64].

Organization can also balance their innovation activities with low levels of exploration and exploitation, but that does not lead them to be ambidextrous organizations. Therefore, we have conceptualized IA as a combined dimension that is measured by multiplying exploratory and exploitative innovation. The simultaneous and complementary relationship of these dimensions is pointed out in this study [90].

We have assessed the SIMS structure based on Karlsson's [91] innovation management capability evaluation questions. This framework covers the six dimensions of the ISO 56002 standard, such as leadership, context, planning, support, and evaluation. We have measured SIMSs with 26 items.

Control variables such as firm size and firm age have been taken into consideration in this study. Larger firms with more resources can allocate these resources to innovation activities, and younger firms, often more entrepreneurial, tend to focus more on exploratory activities [87]. Therefore, both firm size [92] and firm age [93] can influence innovation performance. Firm size is measured as a natural logarithm of the number of employees in the company in this study. Firm age was measured as the natural logarithm of the number of years from the establishment of the company until the survey date.

4.2. Sample and Data Collection

The data for this study were collected from manufacturing companies that are among Turkey's top 500 companies in terms of R&D expenditures [94]. Because manufacturing and service companies appear to have different approaches to innovation processes, we have selected manufacturing companies for data collection to eliminate the potential bias that service sector companies can cause [95].

The survey was initially developed in English, and then it was translated into Turkish by a professional with 25 years of R&D experience. To ensure that the Turkish expressions in the survey were understood in the same way as the originals, two managers who were experts in English and had 25 years of R&D industry experience were asked to review first the Turkish version and then the original English version. Changes were made to the Turkish survey based on the feedback received. Subsequently, an academican with experience in this field reviewed the final version of the survey, and further adjustments were made based on the feedback received. Firstly, a pilot study was conducted between March 2022 and April 2022. The survey was sent to 26 managers, and a pre-test was conducted based on the feedback received from this pilot study to optimize the survey. In the second stage, a survey covering managers in 264 firms was conducted online between May 2022 and January 2023. The survey was sent to innovation managers, R&D managers, and top-level executives. A total of 178 responses were received for the survey, which corresponds to a 67% participation rate. Table 1 summarizes the profiles of the sample.

Table 1. Profile of sample.

Characteristics	Number of Firms	Percentage
Firm Size		
Medium	32	16%
Large	146	82%
Fim Age		
<5 years	1	1%
5–10 years	6	4%
>10 years	171	95%
Position		
Manager	54	30%
Senior manager	94	53%
Executives	30	17%

Source(s): Authors' own construction.

4.3. Common Method Bias

As the data are collected from a single source, common method bias (CMB) may affect the relationships between the constructs [96], and it is a potential threat to the validity of the study [97]. To mitigate CMB, the surveys were administered anonymously. Respondents were clearly informed of the defined objectives, validated scales were employed, and the items were presented in a random order [96]. Furthermore, we assessed the impact of CMB using the Harman one-factor test [98]. By using the Harman one-factor test, we found that common method variance does not pose a serious problem in our study because several factors with eigenvalues greater than 1 were identified, explaining 74.28% of the total variance, and because no factor accounts for almost all the variance (i.e., highest single variance extracted was 36.20%). Therefore, we concluded that common method bias was not a serious problem for this study.

5. Analyses and Results

5.1. Analysis

The measurement model and structural model were assessed by the Partial Least Squares (PLS) method using SmartPLS 4.1.0.2 software [99]. We used SmartPLS 4 software because it is easy to use and needs little technical expertise about the method [100]. Compared with other approaches, PLS-SEM provides greater flexibility in modeling and data requirements (e.g., small samples and non-normally distributed data). PLS-SEM is

more effective in handling regression-based structural equation modeling (SEM) using a principal component-based estimation approach [101] compared to covariance-based SEM. We used the PLS-SEM method because of its ability to accommodate various data distributions without imposing restrictive assumptions [100]. PLS is particularly favored due to its flexibility in handling both reflective and formative constructs [101]. Additionally, PLS-SEM is well suited for analyzing complex models with both direct and indirect relationships [102], especially when the sample size is relatively small, as in this study [100]. This approach is highly suitable for empirical research in the management and marketing fields [103]. The analysis followed a rigorous two-stage process: first, the measurement model was assessed to ensure the construct's reliability and validity; second, the structural model was evaluated to test research hypotheses [100].

5.2. Measurement Model Assessment

All the constructs in this analysis were modeled as reflective constructs. We have assessed the measurement model by evaluating indicator reliability, construct reliability, convergent validity, and discriminant validity [101].

As the first step, we have examined indicator reliability. Loadings are recommended to be above 0.7 [100]. All the indicator loadings were above 0.7 except for two items of IC, which were 0.646 and 0.689. Since they were close to 0.7 and important for the analyses, we decided to keep these two items. Next, Composite Reliability (CR) and Cronbach's alpha values of the reflective model were evaluated. The threshold value for CR [104] and Cronbach's alpha [105] is 0.7. All values were above the threshold value, which confirms the internal consistency reliability of the construct measures. For convergent validity, average variance extracted (AVE) was used. All values were above 0.5, as suggested by Fornell and Larcker [104]. Table 2 shows item loadings, internal consistency reliability, and convergent validity of variables.

Table 2. Reliability analysis and convergent validity of constructs.

Variables	Items	Factor Loading	Cronbach' α	CR	AVE	VIF
SIMS-Context	Context1	0.897 ***	0.934	0.95	0.84	1.096
	Context2	0.907 ***				1.561
	Context3	0.935 ***				1.668
	Context4	0.916 **				1.855
SIMS-Assessment	Assesment1	0.974 ***	0.942	0.97	0.95	1.898
	Assesment2	0.971 ***				1.898
SIMS-Support	Support1	0.877 ***	0.934	0.95	0.79	1.915
	Support2	0.859 ***				1.772
	Support3	0.927 ***				1.855
	Support4	0.906 ***				1.226
	Support5	0.879 ***				1.105
SIMS-Planning	Planning1	0.949 ***	0.962	0.97	0.84	1.449
	Planning2	0.931 ***				1.637
	Planning3	0.948 ***				1.562
SIMS-Process	Process1	0.929 ***	0.937	0.96	0.89	1.702
	Process2	0.943 ***				2.198
	Process3	0.951 ***				1.982
	Process4	0.902 ***				1.862
	Process5	0.932 ***				1.748
	Process6	0.939 ***				2.380
SIMS-Leadership	Leadership1	0.908 ***	0.97	0.98	0.87	1.448
	Leadership2	0.895 ***				1.828
	Leadership3	0.906 ***				2.399
	Leadership4	0.945 ***				1.229
	Leadership5	0.933 ***				1.687
	Leadership6	0.914 ***				1.845

Table 2. Cont.

Variables	Items	Factor Loading	Cronbach' α	CR	AVE	VIF
Innovation Capability	IC1	0.811 ***	0.833	0.88	0.59	1.708
	IC2	0.823 ***				1.853
	IC3	0.842 ***				1.931
	IC4	0.646 ***				1.987
	IC5	0.689 ***				2.079
Innovation Ambidexterity (Exploitative)	Exploitative1	0.708 ***	0.849	0.89	0.63	1.630
	Exploitative2	0.862 ***				2.483
	Exploitative3	0.761 ***				1.827
	Exploitative4	0.731 ***				1.792
	Exploitative5	0.876 ***				2.352
Innovation Ambidexterity (Explorative)	Explorative1	0.892 ***	0.899	0.93	0.77	1.881
	Explorative2	0.921 ***				2.173
	Explorative3	0.851 ***				1.429
	Explorative4	0.832 ***				1.507
Innovation Performance	IP1	0.746 ***	0.883	0.91	0.63	1.959
	IP2	0.822 ***				2.061
	IP3	0.841 ***				2.159
	IP4	0.812 ***				2.100
	IP5	0.768 ***				1.886
	IP6	0.767 ***				1.783

Note(s): ** $p < 0.01$, *** $p < 0.001$. Source(s): Authors' own construction.

Further, discriminant validity was assessed by using Fornell–Larcker [104] and the Heterotrait–Monotrait (HTMT) criteria. Values are shown in Tables 3 and 4. Since the squared root AVE value for each construct was greater than correlations with other model variables [104] and HTMT values were below 0.9 [102], both criteria were met. Therefore, the results indicated that measures are unidimensional and have adequate reliability and validity.

Table 3. Discriminant validity of the constructs—the Fornell and Larcker criterion.

	1	2	3	4	5	6	7	8	9	10
1. IA_Exploitative	0.914									
2. IA_Explorative	0.669	0.973								
3. IC	0.280	0.356	0.891							
4. SIMS_Context	0.599	0.614	0.440	0.918						
5. SIMS_Assesment	0.464	0.511	0.483	0.622	0.943					
6. SIMS_Support	0.551	0.592	0.480	0.705	0.730	0.933				
7. SIMS_Leadership	0.612	0.639	0.454	0.746	0.693	0.752	0.767			
8. SIMS_Planing	0.478	0.496	0.420	0.621	0.683	0.717	0.735	0.791		
9. SIMS_Process	0.480	0.534	0.466	0.671	0.773	0.780	0.721	0.701	0.876	
10. IP	0.464	0.519	0.375	0.561	0.681	0.736	0.731	0.674	0.711	0.794

Source(s): Authors' own construction.

Table 4. Discriminant validity of the constructs—Heterotrait–Monotrait Ratio (HTMT).

	1	2	3	4	5	6	7	8	9	10
1. IA_Exploitative										
2. IA_Explorative	0.722									
3. IC	0.278	0.352								
4. SIMS_Context	0.617	0.632	0.270							
5. SIMS_Assesment	0.459	0.517	0.285	0.623						
6. SIMS_Support	0.564	0.608	0.286	0.714	0.737					
7. SIMS_Leadership	0.618	0.647	0.266	0.743	0.686	0.750				
8. SIMS_Planing	0.483	0.503	0.226	0.622	0.684	0.724	0.732			
9. SIMS_Process	0.469	0.531	0.263	0.662	0.765	0.777	0.701	0.691		
10. IP	0.483	0.545	0.363	0.564	0.531	0.593	0.574	0.521	0.549	

Source(s): Authors' own construction.

5.3. Structural Model Assessment

The structural model was tested by using SmartPLS bootstrapping with 5000 subsamples [106]. Table 5 summarizes the results of the hypotheses testing. The results showed that SIMSs have a positive and significant influence on IC ($\beta = 0.501$, $p < 0.01$) and IA ($\beta = 0.783$, $p < 0.01$), which supports H1 and H3. Secondly, the IC effect on IP has been tested. The outcomes provided evidence of a positive and significant relationship between IC and IP ($\beta = 0.127$; $p < 0.05$), supporting H2. Analysis results also confirmed that SIMSs have a positive and significant influence on IP ($\beta = 0.483$, $p < 0.01$), providing support for H4. Furthermore, results revealed that IA has a positive and significant effect on IP ($\beta = 0.240$, $p < 0.01$), thereby allowing to support H5. Finally, no significant influence of the control variables (firm size, firm age structure) has been found on IP.

Table 5. Structural model results.

Hypotheses	Relationships	Path Value	SD	t-Value	Results
H1	SIMS $\rightarrow$ IC	0.501 **	0.068	7.368	Supported
H2	IC $\rightarrow$ IP	0.127 *	0.073	1.740	Supported
H3	SIMS $\rightarrow$ IA	0.783 **	0.064	12.234	Supported
H4	SIMS $\rightarrow$ IP	0.483 **	0.068	7.103	Supported
H5	IA $\rightarrow$ IP	0.240 **	0.071	3.380	Supported
Size	Size $\rightarrow$ IP	0.012	0.075	0.160	Not Supported
Age	Age $\rightarrow$ IP	−0.03	0.074	−0.405	Not Supported

Note(s): Bootstrapping based on $n = 5.000$ subsamples. * $p < 0.05$, ** $p < 0.01$ (one tailed). Source(s): Authors' own construction.

Collinearity, explanatory power (R^2), the predictive relevance of endogenous variables using Stone–Geisser's Q^2 values, and effect size values f^2 were examined to assess the model fit of the structural model [100].

Before evaluating the structural relationship, we have checked collinearity. We found that all VIF values were below 5 (the maximum VIF value is 2.7), confirming the absence of collinearity [100].

The next step is to evaluate model quality based on its ability to estimate the endogenous constructs [107]. The R^2 indicates the variance of endogenous variables described by the structural model [100]. The R^2 values for endogenous variables were IP (0.710), IC (0.617), and IA (0.279). This confirmed that the exogenous variable has satisfactory explanatory power for the endogenous variables [100].

To assess predictive validity, we have used the Stone–Geisser indicator (Q^2) and the effect size f^2 . Q^2 values higher than zero indicate that exogenous variables have predictive relevance for endogenous variables [106]. Q^2 values were IA (0.612), IC (0.249), and IP (0.597), which confirms that they have adequate predictive relevance. The f^2 value shows the weight of each construct in the model [107]. Effect size values (f^2) higher than 0.35, 0.15, and 0.02 indicate that large, medium, and small effects are present [108]. The results of the structural model assessment are presented in Table 6.

Table 6. R^2 , prediction, and effect size.

Variables	R^2	Q^2
SIMS		
Innovation Ambidexterity	0.617	0.612
Innovation Capability	0.279	0.249
Innovation Performance	0.710	0.597

Source(s): Authors' own construction.

SRMR value was assessed for overall model fit. SRMR value was below 0.08 (0.073), which shows that the model fit is good [109].

6. Discussion and Conclusions

Nowadays, it is becoming more and more common practice to handle innovation activities as a process that involves all divisions of the business, not just the R&D departments. Therefore, companies are adopting different innovation management systems. Some of these companies have achieved very successful innovations, while others have failed. A formalized framework can be created by codifying procedures and routines in a management standard, which is necessary for effective innovation management [10].

Previous studies have studied the relationship between quality management standards and innovation. Some of the findings in these studies support the positive relationship between quality management standards and innovation [110,111], while some have seen these standards as an obstacle to innovation [29,112]. Since quality standards are insufficient for innovation management [113], innovation management standards have begun to be created. Finally, in 2019, the ISO 56002 innovation management standard was established [40].

Research on innovation management standards is very limited. Prior studies in the literature have mostly concentrated on case studies from various industries [5,11–13]. However, very few empirical studies have been conducted. The fundamental goal of this empirical study is to examine the impact of SIMSs on AI and IP, addressing a gap in the literature.

The study results showed that the SIMSs and IC, IA, and IP were positively related. It also confirmed the positive effect of IA and IC on IP.

This study was the first study in the literature to study the relationship between SIMSs and IA empirically. This research also supports dynamic capability theory. SIMSs are considered a dynamic capability as it enables firms to reconfigure resources and adapt to changing market conditions. It encourages analyzing potential opportunities in the market regularly [18,42]. Thus, it allows firms to improve both exploitative and explorative innovation simultaneously, namely IA.

Likewise, it was the first empirical study to confirm a direct positive relationship between SIMSs and IP. As mentioned previously, there are a few empirical studies on SIMSs. In their research, Mir et al. [4] found a positive relationship between SIMSs and business performance, but contrary to expectations, they did not find a direct relationship between SIMSs and IP. In their study, Mir et al. [4] measured SIMSs in UNE 166002 as those with and without certification. As Mir et al. [4] stated in their article, the direction of the causal relationship between SIMSs and IP is open to debate. Again, as stated in the study, companies with a certificate may have already had a high IP before receiving the certificate. This study aimed to measure the application of SIMS principles by companies, regardless of whether they have a certificate or not. We have measured the SIMS structure with a 26-item scale that includes all dimensions of the ISO 56002 standard. For this reason, the results of this study clarified the debate regarding the direction of the causal relationship stated by Mir et al. [4]. This study confirmed a positive relationship between SIMSs and IP, as expected in Mir's [4] study.

We have found a positive relationship between SIMSs and IC, which supports the results of the research conducted by Mir et al. [4] and Mir et al. [40]. While Mir et al. [4] found a positive relationship between SIMSs and IC in their study based on 73 valid surveys, Mir et al. [40] found a positive relationship between the degree of standardization and IC.

The positive relationship between IC and IP identified by this study confirms the findings of researchers such as Mir et al. [4], Mir et al. [40], Sarwar et al. [69], Hurtado-Palomino et al. [52], and Akman and Yilmaz [43].

In this study, a positive relationship between IA and IP was determined. This result supports the findings in the literature [56,64,87,114,115].

This study has important theoretical results. This empirical research is the first study in the literature to confirm the positive relationship between SIMSs and IA. A connection has

been established between the newly developing SIMS literature and the IA literature, which are two different sub-fields in the literature. With this study, a new research area has been opened for future studies and the development of the SIMS literature from this perspective.

Secondly, the positive relationship between SIMSs and IP has been confirmed for the first time. Martinez-Costa et al. [7] found a positive relationship between SIMSs and innovation types in their study. They did not directly measure IP. Mir et al. [4] could not confirm the direct relationship between SIMSs and IP but found a direct relationship between SIMSs and business performance. Giménez Espin et al. [2] found a direct relationship between SIMSs and organizational performance.

Another theoretical contribution is that two previous studies examining the relationship between SIMSs and IC used a sample of companies in Spain [4,40]. This study was conducted in a developing country like Türkiye. This study, which confirms the relationship between SIM and IC in the literature, contributes to the generalization of previous study results.

Fourthly, this study is the first in the literature to measure the SIMS construct using dimensions that include every facet of the ISO 56002 standard principles. The dimensions used by Martinez-Costa et al. [7] and Giménez Espin et al. [2] to measure the SIMS construct, referencing UNE 166002, do not adequately address key concepts of the ISO 56002 standard. These dimensions did not cover SIMS factors like portfolio management, leadership, innovation processes, innovation strategy and policy, and collaborations.

The results of this study are very important for managers. Managers should benefit from innovation management system standards to manage innovation processes with a holistic approach. Companies need to strictly apply the SIMS principles to have sustainable, successful IP and a high degree of IA. For this reason, we anticipate that the implementation of the ISO 56002 innovation management standard by companies will become widespread, just like the development and dissemination of quality standards.

As with other empirical studies, there are some limitations in this study. Taking these limitations into consideration in future studies will allow improvements in this field.

First, the research data for this study were collected from companies in Türkiye. Türkiye is a developing country with its own cultural and contextual factors. Developing countries are less institutionalized than developed countries. They also differ in terms of know-how, information management systems, and technological infrastructure [116]. Differences here may have implications for organizations' SIMS practices and innovation ambidexterity. Thus, the studies conducted in developed and developing countries may have provided different results. To generalize the results of this study, it would be valuable to examine the SIMSs, IA, IC, and IP relationships in the context of a different country. In addition, this research is conducted in manufacturing firms. Since different sectors have different priorities, characteristics, and focuses, their innovation activities, how they innovate, the resources they need to innovate, how they acquire these resources, and the type of innovation they focus on may differ. Therefore, it is possible to reach different findings in studies conducted in different sectors, and such a comparison in future studies may contribute to the literature. In general, the Turkish manufacturing sector has the characteristics of having developed sub-sectors, geographical location advantage, and low labor cost. On the other hand, it has some difficulties where R&D investments are low, access to resources is not easy, and external dependency, especially in technology and energy inputs, is high. Since these unique characteristics may be quite effective on the antecedents, practices, and outcomes of innovation, different results can be achieved in countries and sectors with different characteristics. Second, although the participants were innovation, R&D managers, and executives who were knowledgeable about the company's innovation activities, the data collected reflect the opinion of a single person. In future studies, data collection with more than one participant from a company should be considered.

The conceptual model in this study was examined using a cross-sectional data set. A future longitudinal study could increase the validity of the findings in this study.

In future studies, the effects of control variables such as industrial sector and partnership structure on the results can be assessed.

In this study, IP was measured subjectively. Objective measurement of IP can increase the validity of the outputs obtained in this study. Evaluating the effects of objective measurement of IP can be the focus of future studies.

Additionally, a comparative analysis between companies that apply different IA approaches or types can be performed to investigate the effects of the applied method on the IA and innovation performance relationship.

Finally, in this study, IC was measured as a single dimension. In future studies, factors such as leadership, organizational culture, customer focus, external knowledge utilization, competence management, and the creativity of employees could be included to measure IC. The effect of a multidimensional evaluation of IC should be examined.

This study examines the SIMSs, IA, IC, and IP. Future studies may investigate the potential mediator role of IA and IC. Moreover, other performance outcomes such as financial performance, market performance, and competitive performance may be added to this model. Also, this study does not consider any possible moderators in these relationships. The moderator role of environmental dynamism, technological changes, and organizational culture may improve current knowledge in this area.

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